galaxies can be thought of as single H_2O molecules, while the universe itself is the whole, uniform glass of water. But sooner or later cosmology has to come to grips with the fact that when you examine the cosmos on "finer" scales you discover clumpy structures such as galaxies. And here, once again, we are faced with a puzzle.

If the universe is indeed smooth, uniform, and homogeneous on large scales—features that are supported by observation and that lie at the heart of all cosmological analyses—where could the smaller-scale lumpiness have come from? The staunch believer in standard big bang cosmology can, once again, shrug off this question by appealing to highly favorable and mysteriously tuned conditions in the early universe: "Near the very beginning," such a believer can say, "things were, by and large, smooth and uniform, but not *perfectly* uniform. How conditions got that way, I can't say. That's just how it was back then. Over time, this tiny lumpiness grew, since a lump has greater gravitational pull, being denser than its surroundings, and therefore grabs hold of more nearby material, growing larger still. Ultimately, the lumps got big enough to form stars and galaxies." This would be a convincing story were it not for two deficiencies: the utter lack of an explanation for either the initial overall homogeneity or these important tiny nonuniformities. That's where inflationary cosmology provides gratifying progress. We've already seen that inflation offers an explanation for the large-scale uniformity, and as we'll now learn, the explanatory power of the theory goes even further. According to inflationary cosmology, the initial nonuniformity that ultimately resulted in the formation of stars and galaxies came from *quantum mechanics.*

This magnificent idea arises from an interplay between two seemingly disparate areas of physics: the inflationary expansion of space and the quantum uncertainty principle. The uncertainty principle tells us that there are always trade-offs in how sharply various complementary physical features in the cosmos can be determined. The most familiar example (see Chapter 4) involves matter: the more precisely the position of a particle is determined, the less precisely its velocity can be determined. But the uncertainty principle also applies to fields. By essentially the same reasoning we used in its application to particles, the uncertainty principle implies that the more precisely the value of a field is determined at one location in space, the less precisely its rate of change at